COIMBATORE INSTITUTE OF TECHNOLOGY

(Government Aided Autonomous Institution Affiliated to Anna University, Chennai)

Department Of Electronics And Communication Engineering

23EC422 MICROCONTROLLER LABORATORY

A Project Report On

UNDERGROUND CABLE FAULT DETECTION

Submitted By:

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1. ABSTRACT

Underground cable systems are widely used for reliable power distribution, but fault detection in such systems is difficult due to their hidden nature. This project proposes a simple and cost-effective method to detect cable faults using the resistor drop technique. The system is based on the principle of Ohm's Law, where voltage drop is proportional to resistance and cable length.

In this model, the underground cable is represented using a series of resistors, and faults are simulated using switches at different points. When a fault occurs, the voltage in the circuit changes accordingly. This voltage is measured using a microcontroller with an analog input.

The microcontroller processes the voltage value to determine the approximate distance of the fault from the source. The result is displayed on an LCD for easy monitoring. This system can detect basic faults such as short circuit and open circuit conditions.

The proposed method is low-cost, simple, and suitable for educational purposes. Although it has limitations in real-world applications, it provides a strong foundation for understanding cable fault detection techniques.

2. HARDWARE USED

The physical architecture of the system is designed to provide stable and responsive fault detection using the following components:

- ESP32
- Relay Module
- Current sensor, Voltage sensor, Temperature sensor
- DIP switch
- Resistors
- Connecting Wires

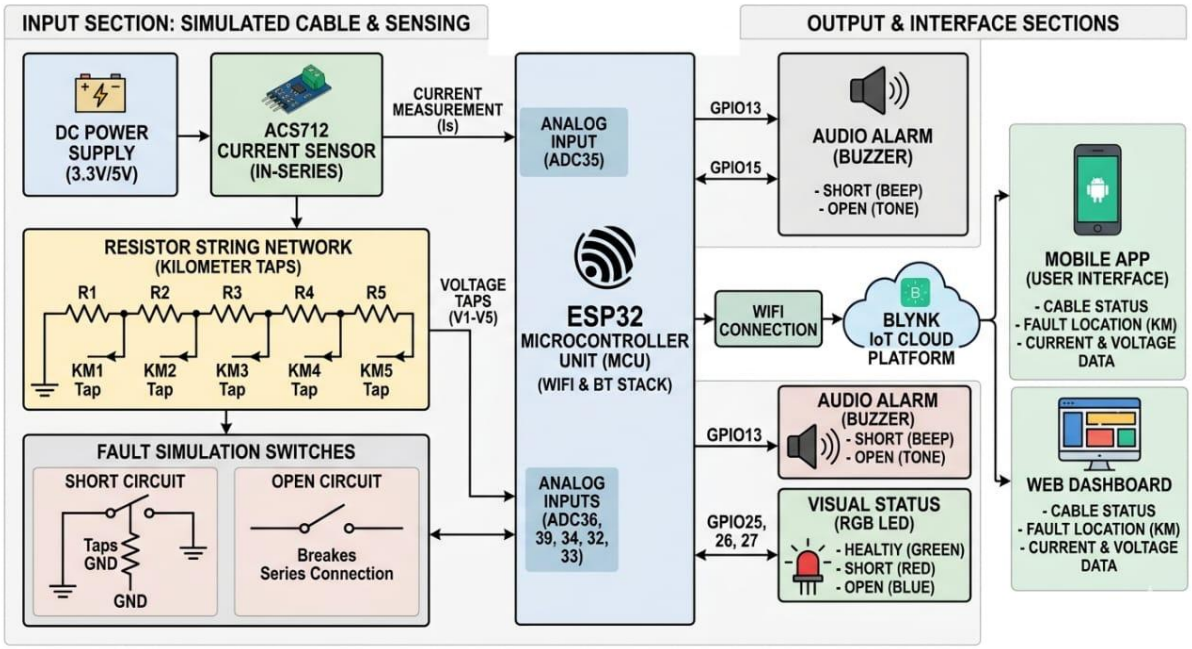
Each component plays a specific role, similar to organs in a body:

- **ESP32** → brain (decision-making & processing data)
- **Voltage, Current & Temperature Sensors** → eyes (monitoring cable condition)
- **Resistors** → cable (representing underground line distance)
- **DIP Switch** → fault creator (simulating cable faults)
- **LCD Display** → mouth (showing fault location & status)
- **Power Supply** → heart (providing energy to system)

3. SOFTWARE USED

- Arduino IDE
- C++ (for coding the microcontroller)
- Serial Monitor (for debugging and output testing)

4. BLOCK DIAGRAM



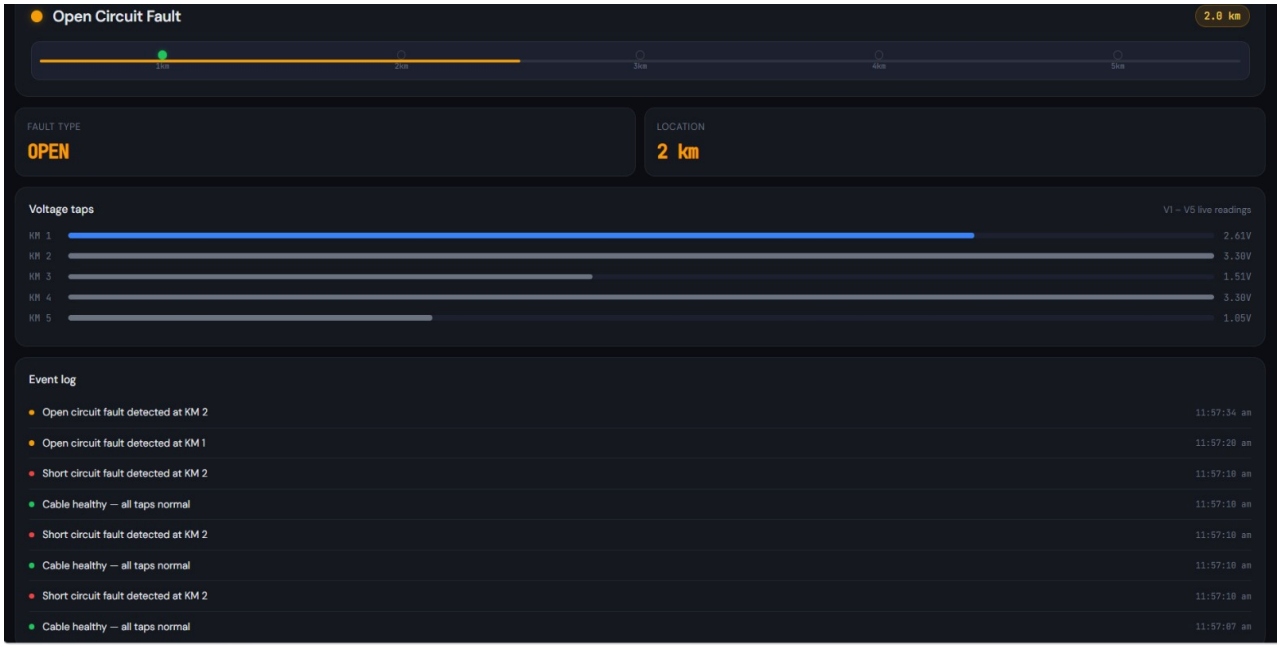
5. ALGORITHM

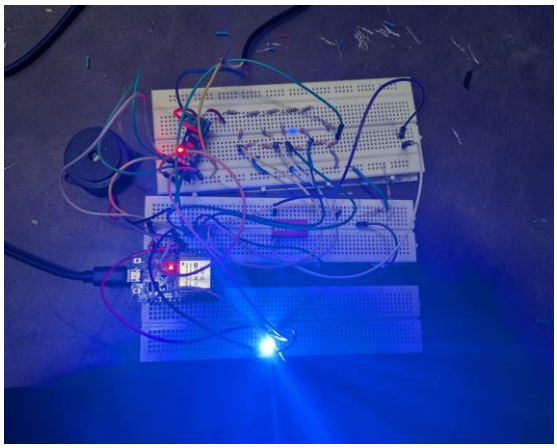
1. Initialize Arduino Uno and all components
2. Read voltage, current, and temperature using sensors
3. Monitor cable condition continuously
4. Compare sensor values with predefined thresholds:
   - If values are normal → Continue monitoring
   - If abnormal variation detected → Fault suspected
5. Identify type of fault:
   - If voltage drops → Open circuit fault
   - If current increases → Short circuit fault
   - If temperature rises → Thermal fault
6. Calculate fault distance using Ohm's Law
7. Confirm fault after short delay (to avoid false detection)
8. Trigger actions:
   - Activate relay to isolate faulty section
   - Display fault type and distance on LCD
9. Repeat process continuously

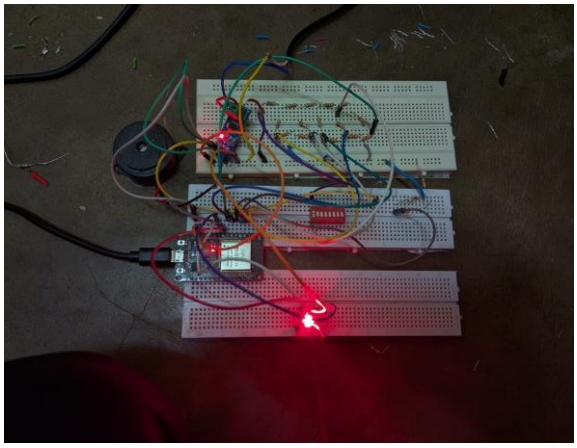
6. RESULT

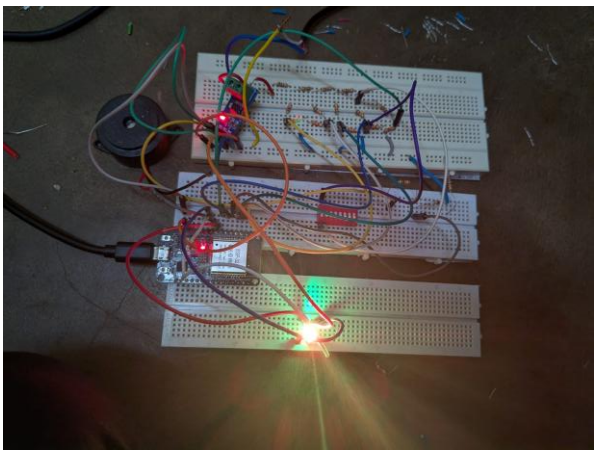
- Monitored cable condition using voltage, current, and temperature sensors.
- Simulated underground cable using resistor network for distance measurement.
- Detected faults based on voltage drop using Ohm's Law.
- Identified different fault types such as open circuit and short circuit.
- Calculated approximate fault distance from the source.
- Isolated faulty section using relay mechanism.

WEB DASH BOARD :









7. CONCLUSION

The proposed system effectively demonstrates a simple and efficient underground cable fault detection mechanism using the resistor drop method. While it is a prototype model, the system successfully identifies fault conditions and estimates the fault location in real time.

The integration of voltage, current, and temperature sensing improves the reliability of fault detection. By applying Ohm's Law, the system is able to calculate the approximate distance of the fault accurately.

This design can be considered a semi-automated system, where detection and indication are handled by the system, while further action can be taken by the user. It provides a strong foundation for developing advanced, fully automated cable monitoring systems in the future.